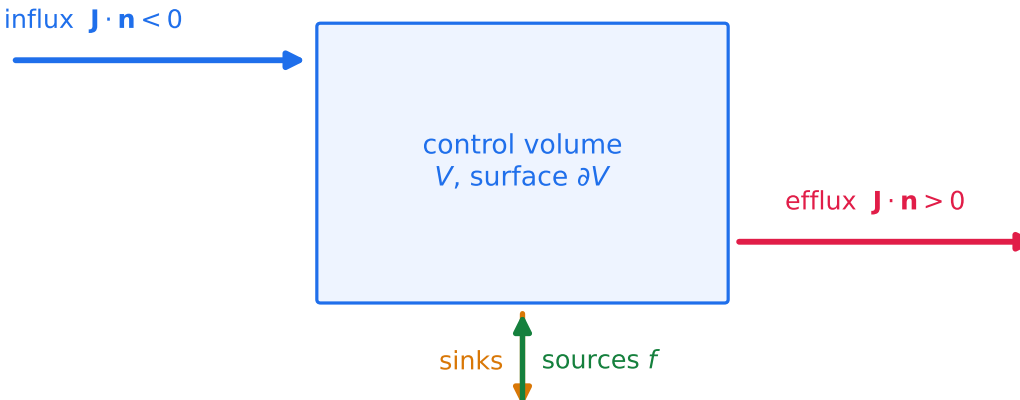
$$\frac{d}{dt}\int_V c\,\mathrm{d}V = -\oint_{\partial V}\mathbf{J}\cdot\mathbf{n}\,\mathrm{d}A + \int_V f\,\mathrm{d}V$$

$\Longrightarrow\ \partial_t c + \nabla\cdot\mathbf{J} = f$ (every PDE in this atlas is an instance of this)



things move, and nothing is lost